



3.0.1 UGC NET CSE | June 2009 | Part 2 | Question: 09



With four programs in memory and with 80% average I/O wait, the CPU utilization is ?

- | | |
|--------|---------|
| A. 60% | B. 70% |
| C. 90% | D. 100% |

ugcnetcse-june2009-paper2

Answer key

3.1

8085 Microprocessor (6)

3.1.1 8085 Microprocessor: UGC NET CSE | December 2014 | Part 3 | Question: 04



Match the following 8085 instructions with the flags :

List – I

List – II

- | | |
|---------|--|
| a. XCHG | i. Only carry flag is affected |
| b. SUB | ii. No flags are affected |
| c. STC | iii. All flags other than carry are affected |
| d. DCR | iv. All flags are affected |

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-iv, b-i, c-iii, d-ii | B. a-iii, b-ii, c-i, d-iv |
| C. a-ii, b-iii, c-i, d-iv | D. a-ii, b-iv, c-i, d-iii |

ugcnetcse-dec2014-paper3 8085-microprocessor non-gatecse

Answer key

3.1.2 8085 Microprocessor: UGC NET CSE | December 2014 | Part 3 | Question: 05



How many times will the following loop be executed ?

LXI B, 0007 H

LOP : DCX B

MOV A, B

ORA C

JNZ LOP

- | | | | |
|-------|-------|-------|-------|
| A. 05 | B. 07 | C. 09 | D. 00 |
|-------|-------|-------|-------|

ugcnetcse-dec2014-paper3 8085-microprocessor non-gatecse

Answer key

3.1.3 8085 Microprocessor: UGC NET CSE | December 2014 | Part 3 | Question: 06



Specify the contents of the accumulator and the status of the S , Z and CY flags when 8085 microprocessor performs addition of $87H$ and $79H$.

- | | | | |
|-------------|-------------|-------------|-------------|
| A. 11,1,1,1 | B. 10,0,1,0 | C. 01,1,0,0 | D. 00,0,1,1 |
|-------------|-------------|-------------|-------------|

ugcnetcse-dec2014-paper3 8085-microprocessor non-gatecse

Answer key

3.1.4 8085 Microprocessor: UGC NET CSE | July 2016 | Part 3 | Question: 3



Which of the following in 8085 microprocessor performs $HL=HL+DE$?

A. DAD D

B. DAD H

C. DAD B

D. DAD SP

ugcnetcse-july2016-paper3 co-and-architecture 8085-microprocessor

Answer key **3.1.5 8085 Microprocessor: UGC NET CSE | November 2017 | Part 2 | Question: 31**

Consider the following program fragment in assembly language:

```

        mov ax, 0h
        mov cx, 0Ah
do loop:
        dec ax
        loop doloop

```

What is the value of ax and cx registers after the completion of the doloop ?A. $ax = FFF5h$ and $cx = 0h$ B. $ax = FFF6h$ and $cx = 0h$ C. $ax = FFF7h$ and $cx = 0Ah$ D. $ax = FFF5h$ and $cx = 0Ah$

ugcnetcse-nov2017-paper2 co-and-architecture 8085-microprocessor

Answer key **3.1.6 8085 Microprocessor: UGC NET CSE | November 2017 | Part 2 | Question: 32**

Consider the following assembly program fragment:

```

stc
mov al, 11010110b
mov cl, 2
rcl al, 3
rol al, 4
shr al, cl
mul cl

```

The contents of the destination register ax (in hexadecimal) and the status of Carry Flag (CF) after the execution of above instructions, are:A. $ax = 003CH$; $CF = 0$ B. $ax = 001EH$; $CF = 0$ C. $ax = 007BH$; $CF = 1$ D. $ax = 00B7H$; $CF = 1$

ugcnetcse-nov2017-paper2 8085-microprocessor non-gatecse

Answer key **3.2****Accumulator (1)****3.2.1 Accumulator: UGC NET CSE | October 2020 | Part 2 | Question: 6**

The following program is stored in memory unit of the basic computer. What is the content of the accumulator after the execution of program? (All location numbers listed below are in hexadecimal).

Location	Instruction
210	CLA
211	ADD 217
212	INC
213	STA 217
214	LDA 218
215	CMA
216	AND 217
217	1234H
218	9CE2H

A. 1002H

B. 2011H

C. 2022H

D. 0215H

ugcnetcse-oct2020-paper2 accumulator assembly-language co-and-architecture

Answer key 

3.3 Addressing Modes (11)

3.3.1 Addressing Modes: UGC NET CSE | August 2016 | Part 3 | Question: 5



In _____ addressing mode, the operands are stored in the memory. The address of the corresponding memory location is given in a register which is specified in the instruction.

- A. Register direct
- B. Register indirect
- C. Base indexed
- D. Displacement

ugcnetcse-aug2016-paper3 co-and-architecture addressing-modes

Answer key 

3.3.2 Addressing Modes: UGC NET CSE | December 2008 | Part 2 | Question: 38



In which addressing mode, the effective address of the operand is generated by adding a constant value to the contents of register ?

- A. absolute mode
- B. immediate mode
- C. indirect mode
- D. index mode

ugcnetcse-dec2008-paper2 co-and-architecture addressing-modes

Answer key 

3.3.3 Addressing Modes: UGC NET CSE | December 2010 | Part 2 | Question: 34



The register or main memory location which contains the effective address of the operand is known as

- A. Pointer
- B. Special location
- C. Indexed register
- D. None of the above

ugcnetcse-dec2010-paper2 co-and-architecture addressing-modes

Answer key 

3.3.4 Addressing Modes: UGC NET CSE | December 2013 | Part 3 | Question: 49



Match the following:

List -I

- a. Indexed addressing
- b. Direct addressing
- c. Register addressing
- d. Base indexed addressing

List-II

- i. is not used when an operand is moved from memory into a register or from a register to memory
- ii. Memory address is computed by adding up two registers plus an (optional) offset
- iii. Addressing memory by giving a register plus a content offset
- iv. can only be used to access global variables whose address is known at compile time

Codes:

- A. a-ii, b-i, c-iv, d-iii
- B. a-ii, b-iv, c-i, d-iii
- C. a-iii, b-iv, c-i, d-ii
- D. a-iii, b-i, c-iv, d-ii

ugcnetcse-dec2013-paper3 co-and-architecture addressing-modes

Answer key 

3.3.5 Addressing Modes: UGC NET CSE | January 2017 | Part 3 | Question: 4



Match the following :

Addressing Mode	Location of operand
a. Implied	i. Registers which are in CPU
b. Immediate	ii. Register specifies the address of the operand
c. Register	iii. Specified in the register
d. Register Indirect	iv. Specified implicitly in the definition of instruction

Codes :

- | | |
|---------------------------|---------------------------|
| A. a-iv, b-iii, c-i, d-ii | B. a-iv, b-i, c-iii, d-ii |
| C. a-iv, b-ii, c-i, d-iii | D. a-iv, b-iii, c-ii, d-i |

ugcnetcse-jan2017-paper3 co-and-architecture addressing-modes match-the-following

Answer key 

3.3.6 Addressing Modes: UGC NET CSE | July 2016 | Part 3 | Question: 5



The ____ addressing mode is similar to register indirect addressing mode, except that an offset is added to the contents of the register. The offset and register are specified in the instruction.

- | | |
|-----------------|-----------------------------------|
| A. Base indexed | B. Base indexed plus displacement |
| C. Indexed | D. Displacement |

ugcnetcse-july2016-paper3 co-and-architecture addressing-modes

Answer key 

3.3.7 Addressing Modes: UGC NET CSE | June 2009 | Part 2 | Question: 33



In which addressing mode the operand is given explicitly in the instruction itself ?

- | | |
|------------------|-------------------|
| A. Absolute mode | B. Immediate mode |
| C. Indirect mode | D. Index mode |

ugcnetcse-june2009-paper2 co-and-architecture addressing-modes easy

Answer key 

3.3.8 Addressing Modes: UGC NET CSE | June 2012 | Part 3 | Question: 27



Identify the addressing modes of the below instructions and match them :

- | | |
|---------|--------------------------|
| (a) ADI | (i) Immediate addressing |
| (b) STA | (ii) Direct Addressing |
| (c) CMA | (iii) Implied Addressing |
| (d) SUB | (iv) Register addressing |

- | | |
|---------------------------|---------------------------|
| A. a-i, b-ii, c-iii, d-iv | B. a-ii, b-i, c-iv, d-iii |
| C. a-iii, b-ii, c-i, d-iv | D. a-iv, b-iii, c-ii, d-i |

ugcnetcse-june2012-paper3 co-and-architecture addressing-modes

Answer key 

3.3.9 Addressing Modes: UGC NET CSE | June 2014 | Part 3 | Question: 40



The advantage of _____ is that it can reference memory without paying the price of having a full memory address in the instruction.

- | | |
|------------------------|---------------------------------|
| A. Direct addressing | B. Indexed addressing |
| C. Register addressing | D. Register Indirect addressing |

ugcnetjune2014iii co-and-architecture addressing-modes

Answer key 

3.3.10 Addressing Modes: UGC NET CSE | June 2019 | Part 2 | Question: 11



Which type of addressing mode, less number of memory references are required?

- A. Immediate B. Implied C. Register D. Indexed

ugcnetcse-june2019-paper2 co-and-architecture addressing-modes

Answer key

3.3.11 Addressing Modes: UGC NET CSE | November 2017 | Part 3 | Question: 4



Which of the following addressing mode is best suited to access elements of an array of contiguous memory locations?

- A. Indexed addressing mode
B. Base Register addressing mode
C. Relative address mode
D. Displacement mode

ugcnetcse-nov2017-paper3 addressing-modes co-and-architecture

Answer key

3.4

Amdahls Law (1)

3.4.1 Amdahls Law: UGC NET CSE | June 2008 | Part 2 | Question: 46



Amdahl's law states that the maximum speedup S achievable by a parallel computer with ' p ' processors is given by :

- A. $S \leq f + (1 - f)/p$
B. $S \leq f/p + (1 - f)$
C. $S \leq 1/[f + (1 - f)/p]$
D. $S \leq 1/[1 - f + f/p]$

ugcnetcse-june2008-paper2 co-and-architecture amdahls-law speedup

Answer key

3.5

Assembly (15)

3.5.1 Assembly: UGC NET CSE | August 2016 | Part 2 | Question: 32



The content of the accumulator after the execution of the following 8085 assembly language program, is

MVI A, 35H
MOV B, A
STC
CMC
RAR
XRA B

- A. 00H B. 35H C. EFH D. 2FH

ugcnetcse-aug2016-paper2 co-and-architecture assembly 8085-microprocessor

Answer key

3.5.2 Assembly: UGC NET CSE | December 2008 | Part 2 | Question: 37



An assembly program contains :

1. imperative and declarative statements
2. imperative and assembler directives
3. imperative and declarative statements as well as assembler directives
4. declarative statements and assembler directives

ugcnetcse-dec2008-paper2 co-and-architecture assembly

Answer key 

3.5.3 Assembly: UGC NET CSE | December 2010 | Part 2 | Question: 31



Object code is the output of _____.

- | | |
|---------------------|--------------------------|
| A. Operating System | B. Compiler or Assembler |
| C. Only Assembler | D. Only Compiler |

ugcnetcse-dec2010-paper2 co-and-architecture assembly

Answer key 

3.5.4 Assembly: UGC NET CSE | December 2010 | Part 2 | Question: 32



'Macro' in an assembly level program is _____.

- | | |
|-----------------------|-----------------------|
| A. Sub program | B. A complete program |
| C. A hardware portion | D. Relative coding |

ugcnetcse-dec2010-paper2 co-and-architecture assembly macros

Answer key 

3.5.5 Assembly: UGC NET CSE | December 2010 | Part 2 | Question: 35



Macro-processors are _____.

- | | | | |
|-------------|-------------|--------------|----------------------|
| A. Hardware | B. Compiler | C. Registers | D. None of the above |
|-------------|-------------|--------------|----------------------|

ugcnetcse-dec2010-paper2 co-and-architecture assembly 8085-microprocessor

Answer key 

3.5.6 Assembly: UGC NET CSE | December 2015 | Part 3 | Question: 2



What will be the output at PORT1 if the following program is executed?

```
MVI B, 82H
MOV A, B
MOV C, A
MVI D, 37H
OUT PORT1
HLT
```

- | | | | |
|--------|--------|--------|--------|
| A. 37H | B. 82H | C. B9H | D. 00H |
|--------|--------|--------|--------|

ugcnetcse-dec2015-paper3 8086 assembly co-and-architecture non-gatecse

Answer key 

3.5.7 Assembly: UGC NET CSE | December 2018 | Part 2 | Question: 15



Consider the following x86 – assembly language instructions:

```
MOV AL, 153
NEG AL
```

The contents of the destination register *AL* (in 8-bit binary notation), the status of Carry Flag (*CF*) and Sign Flag (*SF*) after the execution of above instructions, are

- | | |
|--|--|
| A. <i>AL</i> = 0110 0110; <i>CF</i> = 0; <i>SF</i> = 0 | B. <i>AL</i> = 0110 0111; <i>CF</i> = 0; <i>SF</i> = 1 |
| C. <i>AL</i> = 0110 0110; <i>CF</i> = 1; <i>SF</i> = 1 | D. <i>AL</i> = 0110 0111; <i>CF</i> = 1; <i>SF</i> = 0 |

ugcnetcse-dec2018-paper2 co-and-architecture assembly non-gatecse

Answer key 

3.5.8 Assembly: UGC NET CSE | July 2016 | Part 3 | Question: 4



The Register that stores all interrupt requests is

Interrupt mask register

Interrupt service register

Interrupt request register

Status register

ugcnetcse-july2016-paper3 co-and-architecture assembly

Answer key 

3.5.9 Assembly: UGC NET CSE | July 2018 | Part 2 | Question: 97



Match the items in **List-I** and **List-II** :

List-I

- (a) interrupts which can be delayed when a much highest priority interrupt has occurred
- (b) Unplanned interrupts which occur while executing a program
- (c) Source of interrupt is in phase with the system clock

List-II

- (i) Normal
- (ii) Synchronous
- (iii) Maskable
- (iv) Exception

Code :

- A. (a)-(ii), (b)-(i), (c)-(iv)
- C. (a)-(iii), (b)-(i), (c)-(ii)

- B. (a)-(ii), (b)-(iv), (c)-(iii)
- D. (a)-(iii), (b)-(iv), (c)-(ii)

ugcnetcse-july2018-paper2 co-and-architecture assembly interrupts

Answer key 

3.5.10 Assembly: UGC NET CSE | June 2013 | Part 3 | Question: 41



Which of the following is not an addressing mode?

- A. Register indirect
- C. Relative indexed

- B. Autoincrement
- D. Immediate operand

ugcnetcse-june2013-paper3 co-and-architecture assembly addressing-modes

Answer key 

3.5.11 Assembly: UGC NET CSE | June 2013 | Part 3 | Question: 42



Computers can have instruction formats with

- A. only two address and three address instructions
- B. only one address and two address instructions
- C. only one address, two address and three address instructions
- D. zero address, one address, two address and three address instructions

ugcnetcse-june2013-paper3 co-and-architecture assembly instruction-format

Answer key 

3.5.12 Assembly: UGC NET CSE | June 2013 | Part 3 | Question: 43



What is not a typical program control instruction?

- A. MR B. JMP C. SHL D. TST

ugcnetcse-june2013-paper3 co-and-architecture assembly

Answer key

3.5.13 Assembly: UGC NET CSE | June 2013 | Part 3 | Question: 44



Interrupt which arises from illegal or erroneous use of an instruction or data is

- A. Software interrupt B. Internal interrupt
C. External interrupt D. None of the above

ugcnetcse-june2013-paper3 co-and-architecture assembly interrupts

Answer key

3.5.14 Assembly: UGC NET CSE | June 2015 | Part 2 | Question: 31



Match the following:

a. Foreword Reference Table	i. Assembler directive
b. Mnemonic Table	ii. Uses array data structure
c. Segment Register Table	iii. Contains machine OP code
d. EQU	Uses linked list data structure

- A. $a-ii, b-iii, c-iv, d-i$ B. $a-iii, b-iv, c-ii, d-i$
C. $a-iv, b-i, c-iii, d-ii$ D. $a-iv, b-iii, c-ii, d-ii$

ugcnetcse-june2015-paper2 assembly

Answer key

3.5.15 Assembly: UGC NET CSE | September 2013 | Part 3 | Question: 33



The instruction : MOV CL, [BX] [DI] + 8 represents the ____ addressing mode

- A. Based Relative B. Based Indexed
C. Indexed Relative D. Register Indexed

ugcnetcse-sep2013-paper3 co-and-architecture assembly addressing-modes

Answer key

3.6 Byte Addressable (1)

3.6.1 Byte Addressable: UGC NET CSE | June 2014 | Part 3 | Question: 43



A byte addressable computer has a memory capacity of 2^m Kbytes and can perform 2^n operations. An instruction involving 3 operands and one operator needs a maximum of

- A. $3m$ bits B. $m+n$ bits
C. $3m+n$ bits D. $3m+n+30$ bits

ugcnetjune2014iii co-and-architecture byte-addressable

Answer key

3.7 CO and Architecture (12)

3.7.1 CO and Architecture: UGC NET CSE | December 2008 | Part 2 | Question: 39



Which of the following are Assembler Directives?

- (i) EQU (ii) ORIGIN (iii) START (iv) END

- A. (ii), (iii) and (iv)

- B. (i), (iii) and (iv)
- C. (iii) and (iv)
- D. (i), (ii), (iii) and (iv)

ugcnetcse-dec2008-paper2 co-and-architecture

Answer key 

3.7.2 CO and Architecture: UGC NET CSE | December 2010 | Part 2 | Question: 50



One of the distinguished features of super-computer over other category of computer is

- A. Parallel processing
- B. Highest accuracy level
- C. More speed
- D. More capacity

ugcnetcse-dec2010-paper2 co-and-architecture

Answer key 

3.7.3 CO and Architecture: UGC NET CSE | December 2012 | Part 3 | Question: 27



In an enhancement of a CPU design, the speed of a floating point unit has been increased by 20% and the speed of a fixed point unit has been increased by 10%. What is the overall speed achieved if the ratio of the number of floating point operations to the number of fixed point operations is 2:3 and the floating point operation used to take twice the time taken by the fixed point operation in original design?

- A. 1.62
- B. 1.55
- C. 1.85
- D. 1.285

co-and-architecture ugcnetcse-dec2012-paper3

Answer key 

3.7.4 CO and Architecture: UGC NET CSE | December 2013 | Part 3 | Question: 52



Serial access memories are useful in applications where

- A. Data consists of numbers
- B. Short access time is required
- C. Each stored word is processed differently.
- D. None of these

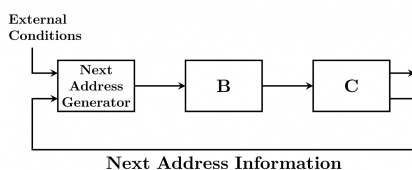
ugcnetcse-dec2013-paper3 co-and-architecture

Answer key 

3.7.5 CO and Architecture: UGC NET CSE | January 2017 | Part 3 | Question: 3



The general configuration of the microprogrammed control unit is given below:



What are blocks B and C in the diagram respectively?

- A. Block address register and cache memory
- B. Control address register and control memory
- C. Branch register and cache memory
- D. Control address register and random access memory

ugcnetcse-jan2017-paper3 co-and-architecture control-unit

Answer key 

3.7.6 CO and Architecture: UGC NET CSE | June 2012 | Part 2 | Question: 3



The branch logic that provides decision making capabilities in the control unit is known as

- A. Controlled transfer
- B. Conditional transfer
- C. Unconditional transfer
- D. None of the above

ugcnetcse-june2012-paper2 co-and-architecture

Answer key

3.7.7 CO and Architecture: UGC NET CSE | June 2012 | Part 2 | Question: 46



A complete micro computer system consists of

- A. Microprocessor
C. Peripheral equipment
B. Memory
D. All of the above

co-and-architecture ugcnetcse-june2012-paper2

Answer key

3.7.8 CO and Architecture: UGC NET CSE | June 2012 | Part 2 | Question: 47



Where does a computer add and compare data?

- A. Hard disk
B. Floppy disk
C. CPU chip
D. Memory chip

co-and-architecture ugcnetcse-june2012-paper2

Answer key

3.7.9 CO and Architecture: UGC NET CSE | June 2012 | Part 3 | Question: 61



Identify the devices given below with their IC numbers :

- (a) USART (i) 8251
(b) Micro Controller (ii) 8051
(c) Interrupt Controller (iii) 8259
(d) DMA Controller (iv) 8257

- A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
B. (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
C. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
D. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

ugcnetcse-june2012-paper3 co-and-architecture

Answer key

3.7.10 CO and Architecture: UGC NET CSE | October 2020 | Part 2 | Question: 44



Which of the following statements with respect to multiprocessor system are true?

- i. Multiprocessor system is controlled by one operating system
ii. In Multiprocessor system, multiple computers are connected by the means of communication lines
iii. Multiprocessor system is classified as multiple instruction stream and multiple data stream system

Choose the correct answer from the options given below:

- A. (i) only
B. (i) and (ii) only
C. (i) and (iii) only
D. (ii) and (iii) only

ugcnetcse-oct2020-paper2 co-and-architecture

3.7.11 CO and Architecture: UGC NET CSE | October 2020 | Part 2 | Question: 81



Given below are two statements:

Statement *I*: Hardwired control unit can be optimized to produce fast mode of operation

Statement *II*: Indirect addressing mode needs two memory reference to fetch operand

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement *I* and Statement *II* are true
B. Both Statement *I* and Statement *II* are false
C. Statement *I* is correct but Statement *II* is false
D. Statement *I* is incorrect but Statement *II* is true

Answer key **3.7.12 CO and Architecture: UGC NET CSE | September 2013 | Part 3 | Question: 31**

Which of the following electronic component is not found in IC's?

- A. Diode B. Resistor C. Transistor D. Inductor

ugcnetcse-sep2013-paper3 co-and-architecture

Answer key **3.8 Cache Memory (5)****3.8.1 Cache Memory: UGC NET CSE | December 2014 | Part 3 | Question: 01**

A hierarchical memory system that uses cache memory has cache access time of 50 nano seconds, main memory access time of 300 nano seconds, 75% of memory requests are for read, hit ratio of 0.8 for read access and the write-through scheme is used. What will be the average access time of the system both for read and write requests ?

- A. 157.5 n.sec. B. 110 n.sec.
C. 75 n.sec. D. 82.5 n.sec.

ugcnetcse-dec2014-paper3 co-and-architecture cache-memory

Answer key **3.8.2 Cache Memory: UGC NET CSE | July 2016 | Part 3 | Question: 6**

In ____ method, the word is written to the block in both the cache and main memory, in parallel

- A. Write through B. Write back
C. Write protected D. Direct mapping

ugcnetcse-july2016-paper3 co-and-architecture cache-memory

Answer key **3.8.3 Cache Memory: UGC NET CSE | July 2018 | Part 2 | Question: 98**

Which of the following mapping is not used for mapping process in cache memory

- A. Associative mapping B. Direct mapping
C. Set-Associative mapping D. Segmented - page mapping

ugcnetcse-july2018-paper2 co-and-architecture cache-memory

Answer key **3.8.4 Cache Memory: UGC NET CSE | June 2012 | Part 2 | Question: 32**

Cached and interleaved memories are ways of speeding up memory access between CPU's and slower RAM. Which memory models are best suited (i.e. improves the performance most) for which programs ?

- I. Cached memory is best suited for small loops.
II. Interleaved memory is best suited for small loops
III. Interleaved memory is best suited for large sequential code.
IV. Cached memory is best suited for large sequential code.
A. I and II are true B. I and III are true
C. IV and II are true D. IV and III are true

ugcnetcse-june2012-paper2 co-and-architecture cache-memory

Answer key **3.8.5 Cache Memory: UGC NET CSE | October 2020 | Part 2 | Question: 5**

Consider a machine with a byte addressable main memory of 2^{16} bytes block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines used with this machine. How many bits will be there in Tag, line and

word field of format of main memory addresses?

- A. 8,5,3 B. 8,6,2 C. 7,5,4 D. 7,6,3

ugcnetcse-oct2020-paper2 co-and-architecture cache-memory

Answer key 

3.9 Cmos (1)

3.9.1 Cmos: UGC NET CSE | July 2018 | Part 2 | Question: 91



CMOS is a computer chip on the motherboard, which is

- A. RAM B. ROM C. EPROM D. Auxillary storage

ugcnetcse-july2018-paper2 co-and-architecture cmos

Answer key 

3.10 Compilation Phases (1)

3.10.1 Compilation Phases: UGC NET CSE | June 2009 | Part 2 | Question: 34



A compiler that runs on one machine and produces code for a different machine is called :

- A. Cross compilation
B. One pass compilation
C. Two pass compilation
D. None of the above

ugcnetcse-june2009-paper2 compiler-design compilation-phases

Answer key 

3.11 Cpu (1)

3.11.1 Cpu: UGC NET CSE | December 2011 | Part 2 | Question: 31



CPU does not perform the operation

- A. Data transfer B. Logic operation
C. Arithmetic operation D. All of the above

ugcnetcse-dec2011-paper2 co-and-architecture cpu

Answer key 

3.12 DMA (1)

3.12.1 DMA: UGC NET CSE | December 2015 | Part 3 | Question: 5



A DMA controller transfers 32-bit words to memory using cycle Stealing. The words are assembled from a device that transmits characters at a rate of 4800 characters per second. The CPU is fetching and executing instructions at an average rate of one million instructions per second. By how much will the CPU be slowed down because of the DMA transfer?

- A. 0.06% B. 0.12% C. 1.2% D. 2.5%

ugcnetcse-dec2015-paper3 co-and-architecture dma

Answer key 

3.13 Data Transfer (1)

3.13.1 Data Transfer: UGC NET CSE | Junet 2015 | Part 3 | Question: 2



Consider a 32-bit microprocessor, with a 16-bit external data bus, driven by an 8 MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate for this microprocessor?

- A. 8×10^6 bytes/sec
C. 16×10^6 bytes/sec
B. 4×10^6 bytes/sec
D. 4×10^9 bytes/sec

ugcnetcse-june2015-paper3 co-and-architecture data-transfer

Answer key

3.14 Dynamic Ram (1)

3.14.1 Dynamic Ram: UGC NET CSE | December 2015 | Part 3 | Question: 4



A dynamic RAM has refresh cycle of 32 times per msec. Each refresh operation requires 100 msec and a memory cycle requires 250 msec. What percentage of memory's total operating time is required for refreshes?

- A. 0.64
B. 0.96
C. 2.00
D. 0.32

ugcnetcse-dec2015-paper3 co-and-architecture dynamic-ram

Answer key

3.15 Flags (1)

3.15.1 Flags: UGC NET CSE | December 2012 | Part 3 | Question: 31



Which of the following flags are set when 'JMP' instruction is executed?

- A. SF and CF
B. AF and CF
C. All flags
D. No flag is set

ugcnetcse-dec2012-paper3 co-and-architecture flags

Answer key

3.16 Hardware Interrupt (1)

3.16.1 Hardware Interrupt: UGC NET CSE | July 2016 | Part 3 | Question: 2



8085 microprocessor has ____ hardware interrupt

- A. 2
B. 3
C. 4
D. 5

ugcnetcse-july2016-paper3 co-and-architecture hardware-interrupt

Answer key

3.17 Instruction Format (1)

3.17.1 Instruction Format: UGC NET CSE | December 2013 | Part 3 | Question: 50



Which of the following is a design criteria for instruction formats?

- A. The size of instructions
B. The number of bits in the address fields
C. The sufficient space in the instruction format to express all the operands desired
D. All of these

ugcnetcse-dec2013-paper3 co-and-architecture instruction-format

Answer key

3.18 Integrated Circuits (1)

3.18.1 Integrated Circuits: UGC NET CSE | December 2011 | Part 2 | Question: 32



A chip having 150 gates will be classified as

- A. SSI B. MSI C. LSI D. VLSI

ugncetcse-dec2011-paper2 co-and-architecture integrated-circuits

Answer key

3.19 Interrupts (3)

3.19.1 Interrupts: UGC NET CSE | December 2015 | Part 3 | Question: 6



A CPU handles interrupt by executing interrupt service subroutine ____

- A. by checking interrupt register after execution of each instruction
B. by checking interrupt register at the end of the fetch cycle
C. whenever an interrupt is registered
D. by checking interrupt register at regular time interval

ugncetcse-dec2015-paper3 interrupts co-and-architecture

Answer key

3.19.2 Interrupts: UGC NET CSE | January 2017 | Part 3 | Question: 1



Which of the following is an interrupt according to temporal relationship with system clock?

- A. Maskable interrupt B. Periodic interrupt
C. Division by zero D. Synchronous interrupt

ugncetcse-jan2017-paper3 interrupts co-and-architecture

Answer key

3.19.3 Interrupts: UGC NET CSE | June 2012 | Part 2 | Question: 37



Interrupts which are initiated by an instruction are

- A. Internal B. External C. Hardware D. Software

co-and-architecture interrupts ugncetcse-june2012-paper2

Answer key

3.20 Language (1)

3.20.1 Language: UGC NET CSE | Junet 2015 | Part 3 | Question: 3



The RST 7 instruction in 8085 microprocessor is equal to

- A. CALL 0010 H B. CALL 0034 H
C. CALL 0038 H D. CALL 003C H

ugncetcse-june2015-paper3 8085-microprocessor assembly language non-gatecse co-and-architecture

Answer key

3.21 Machine Instruction (2)

3.21.1 Machine Instruction: UGC NET CSE | January 2017 | Part 2 | Question: 32



Consider the following assembly language instructions:

```
mov al, 15
mov ah, 15
xor al, al
mov cl, 3
shr ax, cl
```

```
add al, 90H
add ah, 0
```

What is the value in ax register after execution of above instructions?

- A. 0270H B. 0170H C. 01E0H D. 0370H

ugcnetjan2017ii co-and-architecture machine-instruction

Answer key 

3.21.2 Machine Instruction: UGC NET CSE | October 2020 | Part 2 | Question: 75



Arrange the following types of machine in descending order of complexity.

- i. SISD
- ii. MIMD
- iii. SIMD

Choose the correct answer from the options given below:

- A. a, b, c B. c, b, a C. b, c, a D. c, a, b

ugcnetcse-oct2020-paper2 co-and-architecture machine-instruction

Answer key 

3.22 Memory (4)

3.22.1 Memory: UGC NET CSE | June 2019 | Part 2 | Question: 12



Suppose that the register A and the register K have the bit configuration. Only the three leftmost bits of A are compared with memory words because K has 1's in these positions. Because of its organization, this type of memory is uniquely suited to parallel searches by data association. This type of memory is known as

- A. RAM B. ROM
C. content addressable memory D. secondary memory

ugcnetcse-june2019-paper2 co-and-architecture memory

Answer key 

3.22.2 Memory: UGC NET CSE | June 2019 | Part 2 | Question: 19



How many address lines and data lines are required to provide a memory capacity of $16K \times 16$?

- A. 10, 4 B. 16, 16 C. 14, 16 D. 4, 16

ugcnetcse-june2019-paper2 co-and-architecture memory

Answer key 

3.22.3 Memory: UGC NET CSE | September 2013 | Part 3 | Question: 30



Which logic family dissipates the minimum power?

- A. DTL B. TTL C. ECL D. CMOS

ugcnetcse-sep2013-paper3 co-and-architecture memory

Answer key 

3.22.4 Memory: UGC NET CSE | September 2013 | Part 3 | Question: 32



A given memory chip has 14 address and 8 data pins. It has the following number of locations.

- A. 2^8 B. 2^{14} C. 2^6 D. 2^{12}

ugcnetcse-sep2013-paper3 co-and-architecture memory

Answer key 

3.23

Memory Data Transfer (1)

3.23.1 Memory Data Transfer: UGC NET CSE | Junet 2015 | Part 3 | Question: 5



The CPU of a system having 1 MIPS execution rate needs 4 machine cycles on an average for executing an instruction. The fifty percent of the cycles use memory bus. A memory read/write employs one machine cycle. For execution of the programs, the system utilizes 90% of the CPU time. For block data transfer, an IO device is attached to the system while CPU executes the background programs continuously. What is the maximum IO data transfer rate if programmed IO data transfer technique is used?

- A. 500 Kbytes/sec
- B. 2.2 Mbytes/sec
- C. 125 Kbytes/sec
- D. 250 Kbytes/sec

ugcnetcse-june2015-paper3 co-and-architecture memory-data-transfer

Answer key 

3.24

Microprocessors (12)

3.24.1 Microprocessors: UGC NET CSE | August 2016 | Part 3 | Question: 2



8085 microprocessor has ____ bit ALU.

- A. 32
- B. 16
- C. 8
- D. 4

ugcnetcse-aug2016-paper3 co-and-architecture microprocessors 8085-microprocessor

3.24.2 Microprocessors: UGC NET CSE | August 2016 | Part 3 | Question: 4



Which of the following in 8085 microprocessor performs $HL = HL + HL$?

- A. DAD D
- B. DAD H
- C. DAD B
- D. DAD SP

ugcnetcse-aug2016-paper3 co-and-architecture microprocessors 8085-microprocessor

Answer key 

3.24.3 Microprocessors: UGC NET CSE | December 2013 | Part 3 | Question: 54



The essential difference between traps and interrupts is

- A. traps are asynchronous and interrupts are synchronous with the program.
- B. traps are synchronous and interrupts are asynchronous with the program.
- C. traps are synchronous and interrupts are asynchronous with the I/O devices.
- D. None of these

ugcnetcse-dec2013-paper3 microprocessors

Answer key 

3.24.4 Microprocessors: UGC NET CSE | January 2017 | Part 2 | Question: 34



The contents of Register (BL) and Register (AL) of 8085 microprocessor are $49H$ and $3AH$ respectively. The contents of AL , the status of carry flag (CF) and sign flag (SF) after executing ' $SUBAL, BL$ ' assembly language instruction, are

- A. $AL = 0FH$; $CF = 1$; $SF = 1$
- B. $AL = F0H$; $CF = 0$; $SF = 0$
- C. $AL = F1H$; $CF = 1$; $SF = 1$
- D. $AL = 1FH$; $CF = 1$; $SF = 1$

ugcnetjan2017ii microprocessors 8085-microprocessor machine-instruction co-and-architecture

Answer key 

3.24.5 Microprocessors: UGC NET CSE | January 2017 | Part 3 | Question: 5



In 8085 microprocessor, the digit 5 indicates that the microprocessor needs

- A. -5 volts, +5 volts supply
- B. +5 volts supply only
- C. -5 volts supply only
- D. 5 MHz clock

ugcnetcse-jan2017-paper3 co-and-architecture microprocessors

Answer key

3.24.6 Microprocessors: UGC NET CSE | January 2017 | Part 3 | Question: 6



In 8085, which of the following performs: load register pair immediate operation?

- A. LDAX rp
- B. LKLD addr
- C. LXI rp, data
- D. INX rp

ugcnetcse-jan2017-paper3 co-and-architecture microprocessors

Answer key

3.24.7 Microprocessors: UGC NET CSE | July 2018 | Part 2 | Question: 100



In 8085 microprocessor, what is the output of following program?

```
LDA 8000H
MVI B, 30H
ADD B
STA 8001 H
```

- A. Read a number from input port and store it in memory
- B. Read a number from input device with address 8000H and store it in memory location 8001H
- C. Read a number from memory at location 8000H and store it in memory location 8001H
- D. Load A with data from input device with address 8000H and display it on the output device with address 8001H

ugcnetcse-july2018-paper2 co-and-architecture 8085-microprocessor microprocessors

Answer key

3.24.8 Microprocessors: UGC NET CSE | November 2017 | Part 3 | Question: 1



In 8085 microprocessor which of the following flag(s) is (are) affected by an arithmetic operation?

- A. AC flag only
- B. CY flag Only
- C. Z flag Only
- D. AC, CY, Z flags

ugcnetcse-nov2017-paper3 microprocessors 8085-microprocessor co-and-architecture

Answer key

3.24.9 Microprocessors: UGC NET CSE | November 2017 | Part 3 | Question: 2



In 8085 microprocessor the address bus is of _____ bits.

- A. 4
- B. 8
- C. 16
- D. 32

ugcnetcse-nov2017-paper3 microprocessors 8085-microprocessor co-and-architecture

3.24.10 Microprocessors: UGC NET CSE | November 2017 | Part 3 | Question: 3



In the architecture of 8085 microprocessor match the following :

- | | |
|--------------------------------|-------------------------------|
| (a) Processing unit | (i) Interrupt |
| (b) Instruction unit | (ii) General purpose Register |
| (c) Storage and Interface unit | (iii) ALU |
| | (iv) Timing and Control |

Code :

- A. (a)-(iv) (b)-(i) (c)-(ii)
C. (a)-(ii) (b)-(iii) (c)-(i)

- B. (a)-(iii) (b)-(iv) (c)-(ii)
D. (a)-(i) (b)-(ii) (c)-(iv)

ugcnetcse-nov2017-paper3 microprocessors 8085-microprocessor co-and-architecture

Answer key 

3.24.11 Microprocessors: UGC NET CSE | November 2017 | Part 3 | Question: 5



Which of the following is correct statement?

- A. In memory – mapped I/O, the CPU can manipulate I/O data residing in interface registers that are not used to manipulate memory words.
B. The isolated I/O method isolates memory and I/O addresses so that memory address range is not affected by interface address assignment.
C. In asynchronous serial transfer of data the two units share a common clock.
D. In synchronous serial transmission of data the two units have different clocks.

ugcnetcse-nov2017-paper3 microprocessors co-and-architecture

Answer key 

3.24.12 Microprocessors: UGC NET CSE | November 2017 | Part 3 | Question: 6



A micro-instruction format has micro-ops field which is divided into three subfields $F1$, $F2$, $F3$ each having seven distinct micro-operations, conditions field CD for four status bits, branch field BR having four options used in conjunction with address field ADF . The address space is of 128 memory locations. The size of micro-instruction is

- A. 17 bits B. 20 bits C. 24 bits D. 32 bits

ugcnetcse-nov2017-paper3 microprocessors co-and-architecture

Answer key 

3.25 Microprogram Control Unit (1)

3.25.1 Microprogram Control Unit: UGC NET CSE | June 2019 | Part 2 | Question: 14



The fault can be easily diagnosed in the micro-program control unit using diagnostic tools by maintaining the contents of

- A. flags and counters B. registers and counters
C. flags and registers D. flags, registers and counters

ugcnetcse-june2019-paper2 co-and-architecture microprogram-control-unit

Answer key 

3.26 Parallel Computing (1)

3.26.1 Parallel Computing: UGC NET CSE | June 2012 | Part 3 | Question: 5



Suppose that the given application is run on a 64-processor machine and that 70 percent of the application can be parallelized. Then the expected performance improvement using Amdahl's law is

- A. 4.22 B. 3.22 C. 3.32 D. 3.52

parallel-computing ugcnetcse-june2012-paper3

Answer key 

3.27 Parallel Processing (1)

3.27.1 Parallel Processing: UGC NET CSE | Junet 2015 | Part 2 | Question: 46



In the case of parallelization, Amdahl's law states that if P is the proportion of a program that can be made parallel and $(1 - P)$ is the proportion that cannot be parallelized, then the maximum speed-up that can be

achieved by using N processors is:

- A. $\frac{1}{(1-P)+N \cdot P}$ B. $\frac{1}{(N-1)P+P}$ C. $\frac{1}{(1-P)+\frac{P}{N}}$ D. $\frac{1}{P+\frac{(1-P)}{N}}$

ugcnetcse-june2015-paper2 parallel-processing amdahls-law

Answer key 

3.28 Pipelining (6)

3.28.1 Pipelining: UGC NET CSE | December 2004 | Part 2 | Question: 47



The processing speeds of pipeline segments are usually :

- A. Equal B. Unequal C. Greater D. None of these

ugcnetcse-dec2004-paper2 co-and-architecture pipelining

Answer key 

3.28.2 Pipelining: UGC NET CSE | July 2016 | Part 2 | Question: 48



Pipelining improves performance by

- A. decreasing instruction latency B. eliminating data hazards
C. exploiting instruction level parallelism D. decreasing the cache miss rate

ugcnetcse-july2016-paper2 co-and-architecture pipelining

Answer key 

3.28.3 Pipelining: UGC NET CSE | June 2012 | Part 2 | Question: 48



Pipelining strategy is called implement

- A. instruction execution B. instruction prefetch
C. instruction decoding D. instruction manipulation

co-and-architecture pipelining ugcnetcse-june2012-paper2

Answer key 

3.28.4 Pipelining: UGC NET CSE | June 2012 | Part 3 | Question: 3



The concept of pipelining is a most effective in improving performance in the tasks being performed in different stages:

- A. require different amount of time
B. require about the same amount of time
C. require different amount of time with time difference between any two tasks being same
D. require different amount with time difference between any two tasks being different

ugcnetcse-june2012-paper3 co-and-architecture pipelining

Answer key 

3.28.5 Pipelining: UGC NET CSE | October 2020 | Part 2 | Question: 43



Which of the following statements with respect to K -segment pipelining are true?

- i. Maximum speedup that a pipeline can provide is k theoretically
ii. It is impossible to achieve maximum speed up k in k -segment pipeline
iii. All segments in pipeline take same time in computation

Choose the correct answer from the options given below:

- A. (i) and (ii) only
C. (i) and (iii) only

- B. (ii) and (iii) only
D. (i), (ii) and (iii)

ugcnetcse-oct2020-paper2 co-and-architecture pipelining

Answer key 

3.28.6 Pipelining: UGC NET CSE | October 2020 | Part 2 | Question: 7



A non-pipeline system takes 50ns to process a task. The same task can be processed in six-segment pipeline with a clockcycle of 10ns. Determine approximately the speedup ratio of the pipeline for 500 tasks.

- A. 6 B. 4.95 C. 5.7 D. 5.5

ugcnetcse-oct2020-paper2 co-and-architecture pipelining

Answer key 

3.29 Priority Encoder (1)

3.29.1 Priority Encoder: UGC NET CSE | December 2015 | Part 3 | Question: 1



The three outputs $x_1x_2x_3$ from the 8×3 priority encoder are used to provide a vector address of the form $101x_1x_2x_300$. what is the second highest priority vector address in hexadecimal if the vector addresses are starting from the one with the highest priority?

- A. BC B. A4 C. BD D. AC

ugcnetcse-dec2015-paper3 co-and-architecture priority-encoder

Answer key 

3.30 Processor Address Bus (1)

3.30.1 Processor Address Bus: UGC NET CSE | June 2019 | Part 2 | Question: 49



A processor can support a maximum memory of 4 GB where memory is word addressable and a word is 2 bytes. What will be the size of the address bus of the processor?

- A. At least 28 bits B. At least 2 bytes
C. At least 31 bits D. Minimum 4 bytes

ugcnetcse-june2019-paper2 co-and-architecture processor-address-bus

Answer key 

3.31 Register (1)

3.31.1 Register: UGC NET CSE | August 2016 | Part 3 | Question: 3



The register that stores the bits required to mask the interrupts is _____.

- A. Status register B. Interrupt service register
C. Interrupt mask register D. Interrupt request register

ugcnetcse-aug2016-paper3 co-and-architecture register interrupts

Answer key 

3.32 Speedup (2)

3.32.1 Speedup: UGC NET CSE | December 2012 | Part 3 | Question: 49



Let f be the fraction of the computation (in terms of time) that is parallelizable. P the number of processors in the system, and s_p the speed up achievable in comparison with sequential execution – then the s_p can be calculated using the relation:

- A. $\frac{1}{1-f-f/P}$ B. $\frac{P}{P-f(P+1)}$ C. $\frac{1}{1-f+f/P}$ D. $\frac{P}{P+f(P-1)}$

ugcnetcse-dec2012-paper3 co-and-architecture speedup

3.32.2 Speedup: UGC NET CSE | June 2019 | Part 2 | Question: 20

Suppose that a computer program takes 100 seconds of execution time on a computer with multiplication operation responsible for 80 seconds of this time. How much do you have to improve the speed of the multiplication operation if you are asked to execute this program four times faster?

- A. 14 times faster
C. 16 times faster

- B. 15 times faster
D. 17 times faster

ugcnetcse-june2019-paper2 co-and-architecture speedup

Answer Keys

3.0.1	Q-Q	3.1.1	D	3.1.2	B	3.1.3	D	3.1.4	A
3.1.5	B	3.1.6	A	3.2.1	D	3.3.1	Q-Q	3.3.2	D
3.3.3	Q-Q	3.3.4	C	3.3.5	A	3.3.6	D	3.3.7	Q-Q
3.3.8	A	3.3.9	D	3.3.10	X	3.3.11	A	3.4.1	Q-Q
3.5.1	Q-Q	3.5.2	Q-Q	3.5.3	Q-Q	3.5.4	Q-Q	3.5.5	Q-Q
3.5.6	C	3.5.7	Q-Q	3.5.8	C	3.5.9	Q-Q	3.5.10	C
3.5.11	D	3.5.12	C	3.5.13	B	3.5.14	D	3.5.15	B
3.6.1	D	3.7.1	Q-Q	3.7.2	Q-Q	3.7.3	X	3.7.4	D
3.7.5	B	3.7.6	A	3.7.7	D	3.7.8	C	3.7.9	A
3.7.10	C	3.7.11	A	3.7.12	D	3.8.1	A	3.8.2	A
3.8.3	Q-Q	3.8.4	B	3.8.5	A	3.9.1	Q-Q	3.10.1	Q-Q
3.11.1	Q-Q	3.12.1	B	3.13.1	B	3.14.1	D	3.15.1	D
3.16.1	D	3.17.1	D	3.18.1	Q-Q	3.19.1	A	3.19.2	D
3.19.3	D	3.20.1	C	3.21.1	A	3.21.2	C	3.22.1	C
3.22.2	C	3.22.3	D	3.22.4	B	3.23.1	D	3.24.1	Q-Q
3.24.2	Q-Q	3.24.3	B	3.24.4	C	3.24.5	B	3.24.6	C
3.24.7	Q-Q	3.24.8	D	3.24.9	C	3.24.10	B	3.24.11	B
3.24.12	B	3.25.1	D	3.26.1	B	3.27.1	C	3.28.1	Q-Q
3.28.2	C	3.28.3	X	3.28.4	B	3.28.5	A	3.28.6	B
3.29.1	A	3.30.1	C	3.31.1	Q-Q	3.32.1	C	3.32.2	C